

THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NJOMBE REGION
FORM SIX PRE – MOCK EXAMINATION
COMPUTER SCIENCE 1
MARKING GUIDE

SECTION A

1. a) An automated industrial chemical reactor has an "**Emergency Relief Valve**" that only opens if specific conditions are met simultaneously.

Condition A: Reactor temperature exceeds 150°C.

Condition B: Internal pressure exceeds 5 atmospheres.

Condition C: The manual override safety switch is set to ACTIVE.

Required:

- i. Draw the truth table for the above scenario. *(02 Marks)*

Answers

Since the emergency relief valve only opens when all three conditions are met simultaneously, the boolean function is $F = A \cdot B \cdot C$.

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

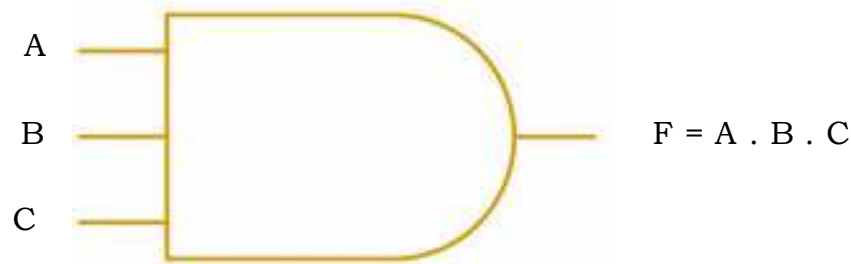
- ii. If the temperature exceeds 150°C ($A=1$) and the safety switch is set to ACTIVE ($C=1$), but the internal pressure is normal ($B=0$), will the relief valve open? Why? *(02 Marks)*

Answer

- No, the valve will not open (Output $F = 0$). *(01 Marks)*
- Reason:** The valve requires all three conditions to be met simultaneously ($A = 1, B = 1, C = 1$). Because the internal pressure is normal ($B = 0$), the logical condition $F = 1 \cdot 0 \cdot 1 = 0$ is not satisfied. *(01 Marks)*

- iii. Draw a logic gate diagram representing this system. (2 Marks)

Answers



- b) Use laws of Boolean algebra to simplify the following expression. (4 Marks)

- i. $ABC + \bar{A}BC + AB\bar{C} + \bar{A}BC$

Answer (02 Marks)

Expression	Law Used
$ABC + \bar{A}BC + AB\bar{C} + \bar{A}BC$	Given
$= ABC + AB\bar{C} + \bar{A}BC$	Idempotent Law
$= AB(C + \bar{C}) + \bar{A}BC$	Distributive Law
$= AB(1) + \bar{A}BC$	Complement Law
$= AB + \bar{A}BC$	Identity Law
$= B(A + \bar{A}C)$	Distributive Law
$= B(A + C)$	Absorption Law

❖ $ABC + \bar{A}BC + AB\bar{C} + \bar{A}BC = B(A + C)$

- i. $ABC + A\bar{B} + AB\bar{C}$ (02 Marks)

Expression	Law Used
$ABC + A\bar{B} + AB\bar{C}$	Given
$= ABC + AB\bar{C} + A\bar{B}$	Commutative law
$= AB(C + \bar{C}) + A\bar{B}$	Distributive law
$= AB.1 + A\bar{B}$	Complementary law
$= AB + A\bar{B}$	Identity law
$= A(B + \bar{B})$	Distributive law
$= A.1$	Complementary law
$= A$	Identity law

❖ $ABC + A\bar{B} + AB\bar{C} = A$

2. A company aims to analyze the performance of a new microprocessor prototype. During testing, they collect the following data from the memory hierarchy, The L1 data cache responds in 4 nanoseconds when it hits. Around 80% of all memory accesses are satisfied by the L1 cache. When an access misses L1, it goes to main memory, which takes 50 nanoseconds to deliver the requested data. The intern notices that some workloads (like image processing)

experience noticeable delays, and they are considering whether to optimize cache performance or memory latency.

- a) Based on the data above, calculate Average Memory Access Time (AMAT) and analyze the performance of this system. What does it suggest about the overall memory behavior? **(5 Marks)**

answer

- ✓ Based on the data above, calculate Average Memory Access Time (AMAT) and analyze the performance of this system. What does it suggest about the overall memory behavior?

Data given (01 marks)

Hit times = 2ns

Hit rates = 90% = 0.9

Miss penalty = 60 ns

Required

AMAT = Hit time + (miss rate * miss penalty) (01 marks)

But,

Miss rate = 1 – Hit rates

Miss rate = 1 – 0.9 = 0.1 (01 marks)

Then,

*AMAT = 2 + (0.1 * 60) = 8 Nanoseconds (01 marks)*

Comment: *In memory behavior this indicates that, when the CPU wants to access data from nearest memory (Cache memory) it has 90% possibility to get data from it and for 10% it misses and find data to other lower memory like RAM and Secondary memory of which the time it takes to find data from lower memory until data is delivered to CPU is 8 nanoseconds (average). (01 marks)*

- b) If software optimization could reduce the miss rate by half, discuss how that would likely impact system responsiveness and why. **(5 Marks)**

Answer

If software optimization could reduce the miss rate by half, discuss how that would likely impact system responsiveness and why.

Now Miss rate = miss rate / 2 = 0.1 / 2 = 0.05 (01 marks)

*AMAT = 2 + (0.05 * 60) = 5 Nanoseconds (02 marks)*

Comment: *When miss rate is reduced by half, The Average memory access time is reduced from 8 ns to 5 ns. This technically indicates system performance increases and in fact hit rates also increases up to 95%. (02 marks)*

3. a) Explain four pillars of object-oriented programming (OOP) with one example each. **(4 marks)**.

Answer (01 marks) @

- ✓ **Encapsulation:** Bundling data and methods in a class.
- ✓ **Abstraction:** Hiding implementation details.
- ✓ **Inheritance:** Creating new classes from existing ones.
- ✓ **Polymorphism:** Same function behaving differently.

- b) Study the following C++ program, which is intended to calculate percentage of a score, then answer the questions that follow.

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int totalScore = 0;
6
7      // Attempting to calculate percentage
8      double percentage = (85 / 100) * 100;
9
10     // Print message
11     cout << "Percentage is: " percentage << endl;
12
13     if (totalScore = 0) {
14         cout << "No score recorded!" << endl
15     }
16
17     return 0;
18 }
```

- i. Identify four errors in the program above. **(4 Marks)**

Answer (01 Marks) @

- **Missing Stream Insertion Operator (<<):** cout << "Percentage is: " percentage requires a << operator before percentage.
- **Missing Semicolon:** Missing ; at the end of cout << "No score recorded!" << endl.
- **Assignment vs. Comparison:** if (totalScore = 0) incorrectly uses the assignment operator (=) instead of the equality comparison operator (==).
- **Integer Division (Logical Error):** 85 / 100 performs integer division resulting in 0. It should be written as 85.0 / 100.0 or (double)85 / 100 to yield a floating-point result.

- ii. Rewrite the program with all the errors corrected. **(2 Marks)**

Answer (02 Marks)

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int totalScore = 0;
6
7      // Fixed integer division by using floating-point literals
8      double percentage = (85.0 / 100.0) * 100;
9
10     // Fixed missing << insertion operator
11     cout << "Percentage is: " << percentage << endl;
12
13     // Fixed equality comparison (==) and added missing semicolon
14     if (totalScore == 0) {
15         cout << "No score recorded!" << endl;
16     }
17
18     return 0;
19 }
```

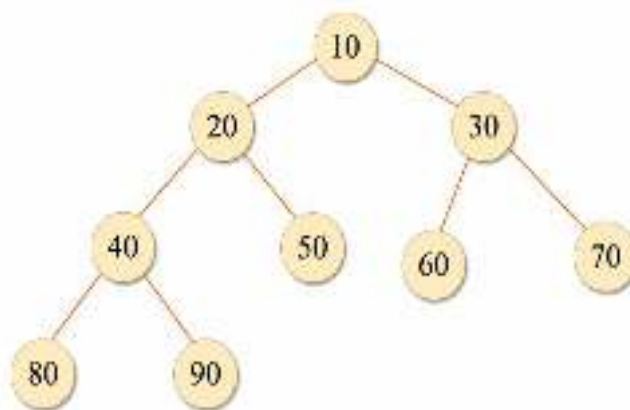
4. a) A file system directory index is represented internally as a binary tree. During a system recovery, the file system manager reconstructs the lost binary tree structure using two logged traversal sequences:

Preorder Traversal: 10, 20, 40, 80, 90, 50, 30, 60, 70

Inorder Traversal: 80, 40, 90, 20, 50, 10, 60, 30, 70

- i. Using the Preorder and Inorder traversal sequences provided, reconstruct and sketch the binary tree. **(03 Marks)**

Answer



- ii. Identify the root node and the maximum depth (height) of the reconstructed tree. **(1 mark)**

Answer

Root & Depth

- Root: **Node 10** **(0.5 Marks)**

- Depth/Height: 3 (assuming root is level 0, max path length is 3 edges: 10 -> 20 -> 40 -> 80) (0.5 Marks)

b) Describe the following operations performed on data structures: (2 marks)

- Insertion
- Deletion
- Searching
- Traversing

Answer: (0.5 Marks)@

- **Insertion:** Adding a new element into a data structure.
- **Deletion:** Removing an existing element from a data structure.
- **Searching:** Finding the location of a particular element.
- **Traversing:** Visiting each element exactly once to process or display it.

c) Explain why data structures are important in computer programming. Give two reasons? (2 marks)

Answer: (01 Marks) @

- Improve program efficiency.
- Reduce memory usage.
- Make searching and sorting easier.
- Simplify data management.
- Improve software performance.

d) Demonstrate how to declare a pointer in C++, and critically explain what happens in memory when the pointer is declared, including how it references or stores the address of a variable. (2 marks)

Answer (2 Marks)

Example:

```
int x = 10;
```

```
int *ptr = &x;
```

- When the pointer is declared, memory is allocated to store the address of variable x. The pointer ptr stores the memory location of x, allowing indirect access or modification of the variable value.

5. a) Apex Manufacturing Ltd is evaluating how to automate its production capabilities across three distinct business units:

Unit A: Produces a single model of aluminum soft drink cans at extremely high volumes (over 500,000 units per day) with zero variation in design.

Unit B: Manufactures custom furniture parts in seasonal batches (e.g., 500 units of Design X this week, 500 units of Design Y next week), where switching product lines takes several hours to retool equipment.

Unit C: Operates an advanced fulfillment facility where automated guided vehicles (AGVs), sorting arms, inventory tracking sensors, and central management software work interdependently in real time.

Questions:

- i. For Unit A and Unit B, identify the most appropriate type of automation system and explain how each type operates in relation to production volume, flexibility, and product variation. **(3 marks)**

Answers

• **Unit A: Fixed Automation** (0.5 Marks)

Explanation: Fixed automation is designed for fixed, high-volume production operations with little to no variation in product design. In Unit A, the sequence of operations is locked directly into the machinery hardware. Because the design never changes and the volume is extremely high (500,000 cans/day), fixed automation yields the highest operational efficiency and lowest unit cost, though it offers zero flexibility to modify the product. (01 Marks)

• **Unit B: Programmable Automation** (0.5 Marks)

Explanation: Programmable automation is used when producing goods in batches with periodic changes in design or specifications. The machinery can be reconfigured or reprogrammed to change the sequence of physical operations for each batch. In Unit B, the several hours required to retool equipment between batches of custom furniture (e.g., Design X to Design Y) is a classic characteristic of programmable automation, balancing high production efficiency with batch-level product variation. (01 Marks)

- ii. Name the type of automation implemented in Unit C. Explain one key advantage and one major vulnerability of deploying this specific architecture in Unit C's fulfillment facility. **(3 marks)**

Answer

- Type of automation implemented in Unit C is **Integrated Automation** (01 Marks)
- **Advantage:** Provides maximum system-wide efficiency, real-time monitoring, and centralized control across disparate devices (sensors, AGVs, management software). (01 Marks)
- **Vulnerability:** High dependency on network infrastructure and central control systems—a disruption or failure at the network or software level can paralyze the entire factory floor (single point of failure), alongside requiring specialized, highly skilled personnel to maintain. (01 Marks)

- b) Describe five real-life applications of automated systems. **(4 marks)**

Answer (01 marks) @

- Automatic washing machines
- Traffic light systems

- Automatic teller machines (ATMs)
- Industrial robots
- Automatic irrigation systems

6. a) Define an Operating System (OS) based on its structural placement within a computer system hierarchy. (2 marks)

Answer (2 Marks)

- An Operating System is a type of system software that stands/acts as an interface between a computer system's hardware and application software/users to facilitate smooth operations.
- b) An Operating System is often compared to an orchestra's conductor, as it coordinates various components to work together harmoniously. Explain three functions Operating System **(3 marks)**

Answers (01 Marks) @

- **Resource Management:** It manages hardware resources like the Central Processing Unit (CPU), memory (RAM), and storage space.
 - **Process Management:** It allocates resources to different programs running simultaneously, ensuring they run smoothly without interference.
 - **Software Interface:** It allows users to install and run application software such as word processors, games, and web browsers.
 - **Hardware Interface:** It acts as a gatekeeper, ensuring that programs interact correctly with the physical hardware.
- c) A software development company, DevCore Tech, is engineering a real-time multiplayer gaming application designed to support thousands of concurrent users. During beta testing, developers noticed that when multiple applications run in the background, the game experiences severe frame rate drops, unexpected system crashes, and memory allocation delays. Additionally, third-party cheat software was able to modify game data directly in RAM because of weak memory isolation.
- i. Identify three key OS design principles that were compromised or inefficiently implemented in the scenario above. **(3 marks)**

Answer (01 Marks) @

- **Security:** Failed to protect memory space from unauthorized modification/malware (e.g., cheat software accessing RAM).
 - **Stability:** The system failed to run smoothly and reliably, causing crashes during high resource load.
 - **Performance:** Inefficient resource utilization created severe lag/frame drops under multitasking conditions.
- ii. Explain how the principle of Stability directly impacts user experience when running resource-intensive application software like DevCore's game. **(2 marks)**

Answers (02 Marks)

- An OS designed for high stability ensures that system crashes and unhandled execution errors are minimized or isolated. This guarantees that application software can run reliably over long periods without system-wide failure, providing a seamless user experience.

7. (a) At the very basic level, a computer system is a device consisting of three components. Mention and discuss these three components. **(3 Marks)**

Answer (01 Marks) @

- ✓ **Central Processing Unit (CPU):** The "brain" of the computer. It interprets and executes instructions through the Control Unit (CU) and the Arithmetic Logic Unit (ALU).
- ✓ **Memory (Primary Storage/RAM):** Holds data and instructions currently in use by the CPU. It is volatile, meaning it loses data when power is cut.
- ✓ **Input/Output (I/O) Devices:** These allow communication between the user and the machine. Input devices (keyboard/mouse) send data to the system; output devices (monitor/printer) display the processed results.

b) A Mr. Jones, a form five student at Mkwawa secondary school intends to buy a computer that he could use while at school as a tool while in the school as well in the hostels. Kindly advise him on the kind and type of a computer system he could buy. Give him the full specification of a good computer. **(4 Marks)**

Answer (01 Marks) @

Recommended Specifications:

- ✓ **Processor:** Intel Core i5 or AMD Ryzen 5 (11th Gen or newer).
- ✓ **RAM:** 8GB or 16GB (for smooth multitasking).
- ✓ **Storage:** 512GB SSD (Solid State Drives are faster and more durable than HDDs).
- ✓ **Display:** 14-inch Full HD (1080p) for portability and clarity.
- ✓ **Battery:** At least 6-8 hours of backup.

c) Explain two difference between the terms 'computer' and 'computer system'. **(3 Marks)**

Answer (1.5 Marks) @

Computer	Computer System
A computer is an electronic device that processes data according to instructions.	A computer system is the complete setup that includes the computer and all the components needed for it to function.
It mainly refers to the physical machine (hardware), such as the CPU, memory, and storage.	It includes hardware, software, data, users, and peripheral devices working together.
Its main function is to accept	Its main function is to provide a

input, process data, store information, and produce output.	complete environment for performing computing tasks efficiently.
<i>Example: A desktop PC or a laptop.</i>	<i>Example: A desktop PC with its operating system, application software, printer, keyboard, mouse, and the user operating it.</i>

SECTION B (30 Marks)

Answer two (2) questions in this section. Each question carries fifteen (15) marks.

8. Network model defines the protocols for network communications. Explain the functions of each open system interconnect model layer in network communication.

Answer

Introduction (01 Mark)

Functions for each layer of the OSI model: (2 Marks) @

- **Physical Layer (Layer 1)** :Transmits raw bits (0s and 1s) over physical media (cables, fiber optics, Wi-Fi). It converts frames from the data link layer into electrical, optical, or radio signals.
- **Data Link Layer (Layer 2)**: Packages packets into frames, handles physical (MAC) addressing, error detection/correction, and flow control on the local network. Sublayers include MAC (Media Access Control) and LLC (Logical Link Control). **Examples:** Framing, Node-to-Node Delivery, Switching.
- **Network Layer (Layer 3)**: Determines the best physical path (routing) for data, manages logical addressing (IP addresses), and breaks down segments into packets. **Examples:** Routing, Packet Forwarding, Logical Addressing, Internetworking.
- **Transport Layer (Layer 4)** :Manages end-to-end communication, breaking data into segments, providing error checking (TCP), flow control, and sequencing.**Examples:** End-to-End Control, Segmentation, Segmentation and Reassembly (SAR).
- **Session Layer (Layer 5)** : Establishes, manages, maintains, and terminates connections (sessions) between applications on different devices. It handles authentication and reconnection. **Examples:** Dialogue Control, Connection Management, Synchronization.
- **Presentation Layer (Layer 6)**: Formats, compresses, and encrypts data to ensure that it is readable by the application layer.**Examples:** Data Formatting, Encryption/Decryption, Translation.
- **Application Layer (Layer 7)** : Interacts with software applications to provide network services (e.g., HTTP, FTP, SMTP). **Examples:** User Interface, Network Application, Software Layer.

Conclusion (01 Marks)

9. As part of the government's "SmartNation Project," junior data analysts are being deployed across public and private industries to implement data-driven decision-making. Identify four sectors where data science is applied, and explain one key operational problem data science helps to solve within each sector.

Answer

Introduction: (02 marks)

Points (03 marks) @

- **Health:** Disease Outbreak Prediction & Diagnostic Accuracy
- **Business & Finance:** Financial Fraud & Supply Chain Inefficiency
- **Education:** Student Dropout Rates & Unidentified Learning Difficulties
- **Agriculture:** Unpredictable Crop Yields & Resource Misallocation

Conclusion (01 marks)

10. "ST. Jude Academy," a growing secondary school, currently tracks student admissions, fee payments, examination marks, and library book loans through separate desktop spreadsheets maintained by individual staff members. This setup has led to widespread problems, including duplicate student profiles, inconsistent grade reports, loss of historical academic records, and delayed report card generation. To resolve these issues, the administration is deploying a centralized Database Management System (DBMS). Using the scenario above, write an essay explaining six key significance of a Database Management System (DBMS) in managing institutional data. **(15 Marks)**

Answer

Introduction: (02 marks)

Points (02 marks) @

- **Reduced Data Redundancy & Elimination of Inconsistencies:** Centralized storage prevents duplicate files and ensures single-point updates.
- **Enhanced Data Security & Access Control:** Role-based permissions protect sensitive operational data from unauthorized access.
- **Improved Data Integrity & Accuracy:** Enforces validation constraints (e.g., preventing negative stock or invalid IDs).
- **Efficient Data Retrieval & Reporting:** Uses query languages (SQL) to fetch complex records in seconds instead of manual searching.
- **Data Sharing & Multi-User Access:** Enables multiple departments or branches to access updated records simultaneously without conflicts.
- **Reliable Backup & Disaster Recovery:** Automated logging and backup protocols prevent catastrophic data loss during system crashes.

Conclusion (01 marks)

